

Substring Order I

[TASK](#) | [SUBMIT](#) | [RESULTS](#) | [ANALYSIS](#) | [STATISTICS](#) | [TESTS](#) | [QUEUE](#)

Time limit: 1.00 s Memory limit: 512 MB

You are given a string of length n . If all of its distinct substrings are ordered lexicographically, what is the k th smallest of them?

Input

The first input line has a string of length n that consists of characters a–z.

The second input line has an integer k .

Output

Print the k th smallest distinct substring in lexicographical order.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq k \leq \frac{n(n+1)}{2}$
- It is guaranteed that k does not exceed the number of distinct substrings.